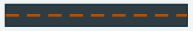
REFERENCE / 01

Parts catalog

Finished dimensions in inches. Six cut textile pieces; thread additional. Bag is reference-only.

#1 Main outer body / Qty 1 34 wide; 7 base / 10 max high		Moisture-wicking polyester-spandex compression knit
#2a Lower interior support / Qty 1 10 x 3 max		Soft polyester-spandex knit
#2b Upper interior support / Qty 1 10 x 3 max		Soft polyester-spandex knit
#3 Identical top and bottom bands / Qty 2 34 x 2 finished; cut 34.5 x 4.5		Smooth matte polyester-spandex with firm recovery
#4 Full-height elastic back insert / Qty 1 2 x 11 finished; cut 2.5 x 11.5		Soft elastic knit; greater stretch than main body
#6 Sewing thread / Qty as needed Length as needed; width / height not applicable		Black polyester needle thread; soft textured polyester looper thread

REFERENCE BAG

Horizontal 10 x 6 in envelope. Not part of product. Left flange is diagram orientation, not a confirmed anatomical side.

CONSTRUCTION

Slip-on. Open body and bands join a 11-inch full-height back insert. No separate closure or bottom binding.

Cut lines, folds and allowances

Finished sizes are targets. The master layout shows both cut and finished boundaries; not a 1:1 print pattern.

PART / FINISHED TARGET	CUTTING RULE
#1 34 wide; 7 base / 10 max high	Offset all joining edges by 0.25 in; preserve the shaped zone.
#2a 10 x 3 max	Add 0.25 in only at turned attachment perimeter; none at free overedge.
#2b 10 x 3 max	Same allowances as #2a, reflected vertically; both cutouts stay left.
#3 34 x 2 finished; cut 34.5 x 4.5	Cut two 34.5 x 4.5 in strips; keep short ends open.
#4 2 x 11 finished; cut 2.5 x 11.5	Cut 2.5 x 11.5 in; side seams plus short-edge hems.

FOLDED BAND CALCULATION

4.5 / 2 - 0.25 = 2 in finished projection. The short-end allowances join #4, not each other.

LINE LEGEND

Pale outline = cut. Blue dashed = finished seam / fold-under line. Orange dashed = fold or basting. Red = leave open.

Equipment, needles and thread

Manufacturer guidance informs choices; the actual machines and fabric stack determine final setup.

MACHINES

Serger: structural four-thread join; soft three-thread free-edge finish.

Coverstitch: back-insert hems and selected allowance finishing.

Standard machine: stretch-capable zigzag for pocket perimeter.

Rotary cutter, clips, removable markers, blunt needle, gauge and curved fixture.

THREAD ROLES

Polyester needle thread, sized to the fabric and machine.

Soft textured polyester in loopers where compatible; check next-to-skin feel.

Use matching black for structural seams; decoration must not restrict stretch.

Needle and looper threads are not interchangeable without testing.

NEEDLE / FABRIC

Use the machine-specified needle system and a knit-compatible point.

Domestic stretch needles are not a universal serger substitute.

Keep stretch around the wearer; test doubled #3 separately.

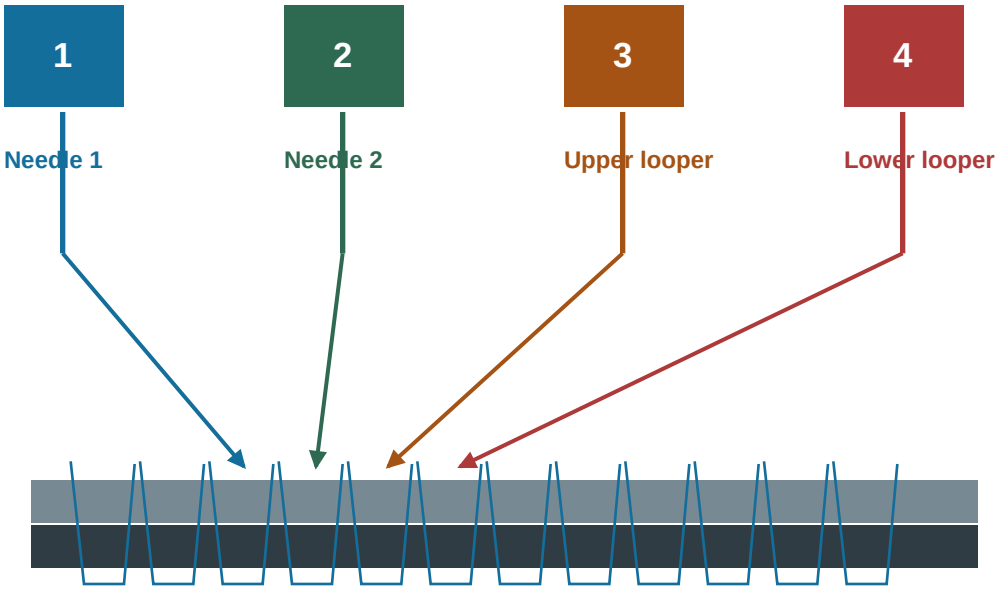
Follow supplier care and heat limits; no branded fabric performance assumed.

COUPON FIRST

Record machine model, needle system/size, thread positions, stitch length/width, differential feed and foot pressure. Begin with manufacturer knit settings. Change one variable at a time; retain an approved sample for each layer stack. Sources: AE, COATS, SCHMETZ.

Serger threading and seam anatomy

Thread-role schematic, not a machine-specific threading route. Follow the actual manual for threading order.



Two plies / schematic enlargement - not to scale

Four-thread construction: two needle positions plus two looper positions. Confirm the machine formation before using an ISO stitch code.

Three-thread edge finish: use the machine-approved needle configuration for a soft free edge. This finish does not replace the structural join.

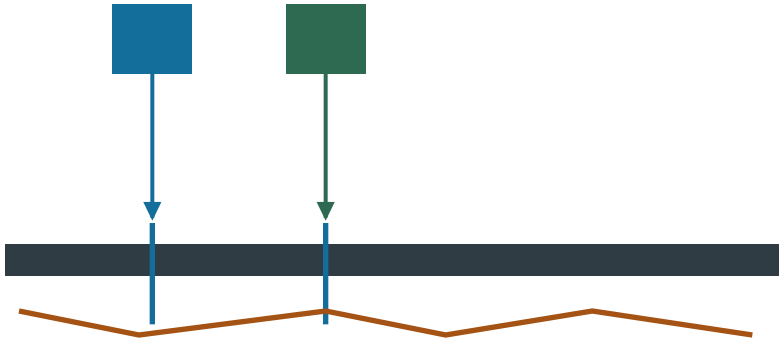
FEED, BALANCE AND END SECURITY

Guide without pulling. Differential feed is adjusted on the coupon to remove waves without gathering the garment. Avoid needle cuts, exposed loops and excessive tension that limits extension. Secure chain tails through the seam with a blunt needle where the machine method permits; inspect for unraveling. Production threads are black; colors here identify roles. Sources: AE, COATS, PEGASUS.

Coverstitch and pocket zigzag

Role diagrams and seam choices. Thread according to the actual machine manual; retain settings with the sample.

COVERSTITCH / TWO NEEDLES + LOOPER



Needle threads form face rows; looper thread covers the underside. Check tunneling and extension. Secure ends by the machine-approved method; simply clipping tails may allow unraveling.

STANDARD MACHINE / NEEDLE + BOBBIN



Turned pocket edge; zigzag stays in allowance

Use the qualified stretch zigzag along the turned attachment perimeter. Free insertion edges remain open. Set width and length on coupons; do not substitute a rigid straight stitch.

COMMON ERRORS

Tunneling: review thread tension, needle spacing and fabric support. Skipped stitches: confirm needle system, point, condition and threading. Cracking under stretch: review seam formation and thread consumption. Do not assume a domestic serger flatlock is an industrial flatseam. Sources: COATS, SCHMETZ, PEGASUS.

REFERENCE / 06

Operation chart and troubleshooting

Prototype choices; record successful settings instead of copying universal tension numbers.

CUT	**Cut, mark and orient**	rotary cutter / shears / none
EDGE	**Finish free pocket edges**	serger / soft three-thread overedge; confirm formation
POCKET	**Attach turned pocket perimeter**	standard sewing machine / stretch-capable zigzag; coupon-qualified
JOIN	**Structural join**	serger / four-thread overlock; confirm formation
HEM	**Finish insert ends / selected allowances**	coverstitch / two-needle coverseam; confirm formation
FOLD	**Fold matching edge bands**	clips / hand positioning / none; removable holding basting only
EASE	**Ease expansion into straight bands**	hand basting then serger / temporary basting then qualified JOIN

WAVES / BULK

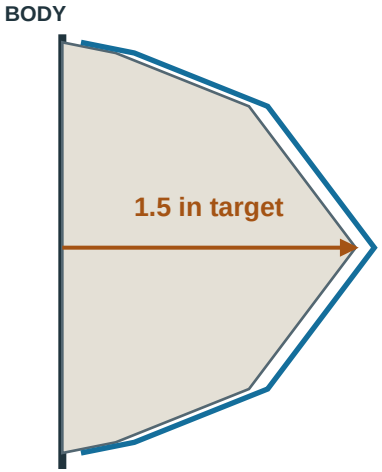
Guide without stretching. Review differential feed, pressure, seam allowance and thread balance on coupons. Do not fix a mismatch by pulling every layer.

STOP / REVIEW

Stop for needle cuts, broken thread, blocked access or hard folds. Record the cause and review connected materials, geometry and instructions. Sources: AE, COATS.

Pouch expansion and easing

The flat maximum height is a design dimension. Outward depth is a separate, unverified physical requirement.



Side section / schematic - not to scale

Measure perpendicular to the local body plane. Flat extra fabric is not a depth guarantee.

REVISED BODY EDGE

Base 7 in; max 10 in. Extra 1.5 in above and below across the 12-inch zone.

Each body seam line measures 34.639 in against a 34-inch band seam line. Local excess: 0.639 in (5.3% of the straight zone).

Compute and baste the excess locally. Do not assume stretching, trimming or a flat measurement guarantees capacity.

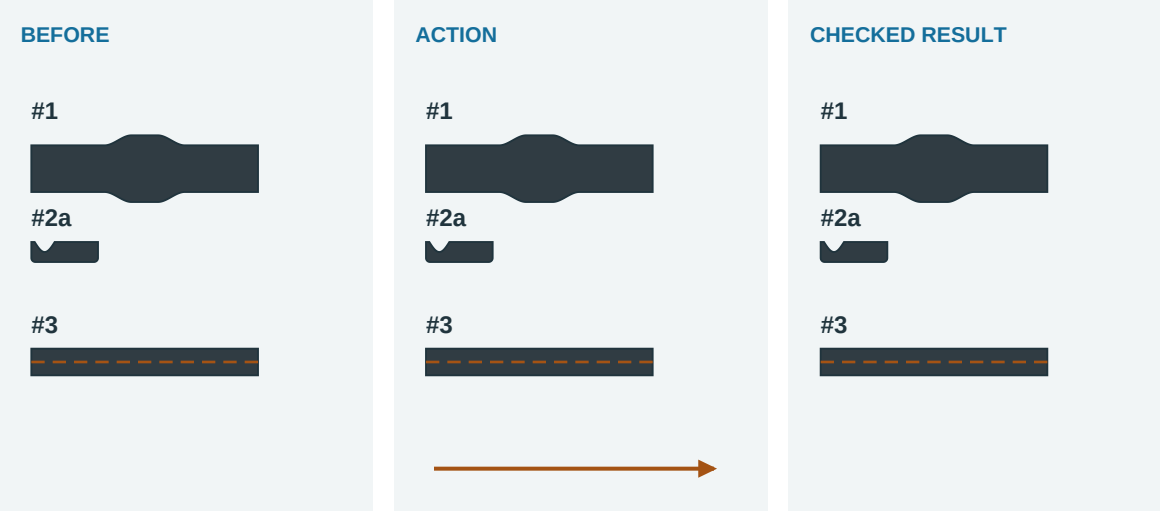
DEPTH CHECK / BEFORE PERMANENT SEWING

Use a curved body-representative fixture and smooth mock-up or gauge. Measure perpendicular to the local body plane at intermediate depth and at 1.5 in outward projection. Check flange clearance, outlet access, insertion, removal, retention, recovery and band flatness. Do not sew with the reference bag in place. A failure stops the affected operation and triggers review of #1, #2a, #2b, #3 and material stretch together. No clinical pressure or bag capacity is established by this check.

ASSEMBLY / STEP 01

Prepare the fabrics

Parts #1, #2a, #2b, #3, #4 | CUT: rotary cutter / shears



WHAT TO DO

1. Record supplier, lot, fiber content and care instructions for every material.
2. Condition or prewash according to supplier instructions; let fabric relax flat before measuring.
3. Mark right/wrong sides. Orient greatest usable stretch around the wearer, including across the narrow back insert.
4. Compare recovery of single body fabric, doubled band fabric and the insert. Keep test offcuts.

SEWING OPERATION

Cut, mark and orient - none

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- No distorted or damaged fabric enters cutting.
- The insert stretches more than the body; doubled bands are checked separately.

AVOID THIS

Do not infer stretch from material names or pull the fabric while cutting.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

ASSEMBLY / STEP 02

Qualify each seam on coupons

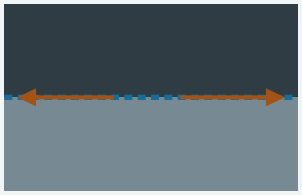
Parts #1, #2a, #2b, #3, #4, #6 | JOIN: serger

BEFORE



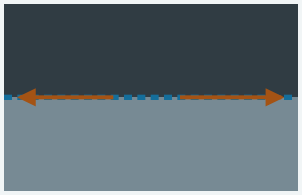
Match layers

ACTION



Extend / release

CHECKED RESULT



Inspect and record

WHAT TO DO

1. Thread using the machine manual; record needle system, size and thread in each position.
2. Sew body-plus-insert, folded-band-plus-body and pocket-on-body coupons using the same layers as the garment.
3. Start with the manufacturer knit setting. Change only one setting at a time until feeding is flat.
4. Extend through the intended assembly use, release, and inspect. Keep the approved coupon and settings record.

SEWING OPERATION

Structural join - four-thread overlock; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- No thread cracking, skipped stitches, needle cuts or persistent waves.
- Retest after thread, needle, material or layer changes.

AVOID THIS

No universal tension number or stitch length can substitute for these coupons.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

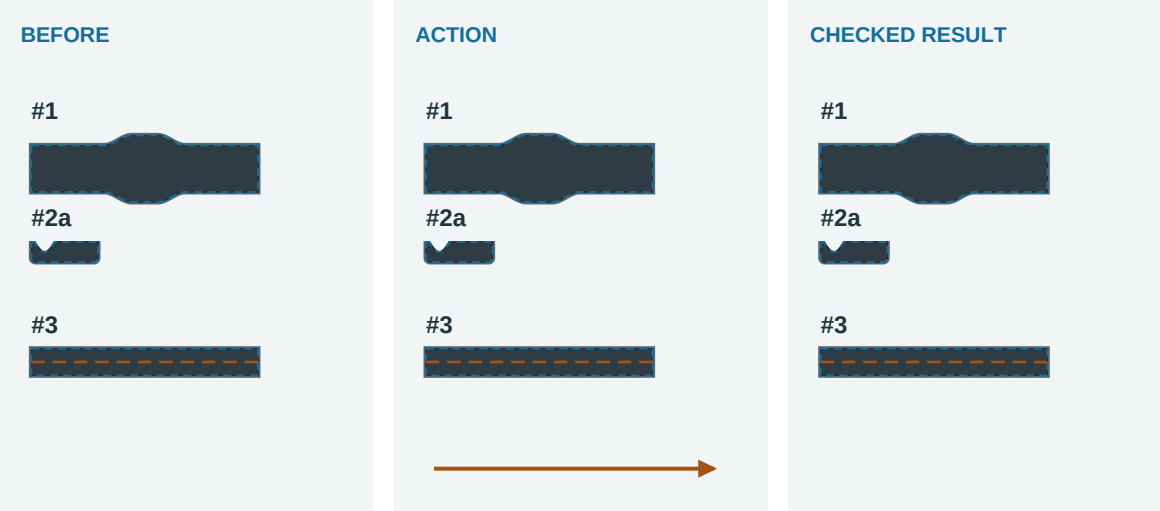


Layer schematic

ASSEMBLY / STEP 03

Cut and mark six textile pieces

Parts #1, #2a, #2b, #3, #4 | CUT: rotary cutter / shears



WHAT TO DO

1. Use the catalog cutting outlines, not the finished seam-line silhouettes. Cut two identical open #3 strips.
2. Transfer center-front, pouch-zone boundaries, band midpoints and back-insert section marks into allowances.
3. Keep both support cutouts on the diagram left when viewed from inside. Mirror vertically for the upper support.
4. Mark free insertion edges in removable chalk. Do not snip notches into the finished seam line.

SEWING OPERATION

Cut, mark and orient - none

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Six pieces counted; cut lines and finished lines remain distinguishable.

- No added hardware or closed band loops.

AVOID THIS

A finished 2-inch band needs a 4.5-inch cut strip before folding.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

ASSEMBLY / STEP 04

Finish the lower free edge

Parts #2a, #6 | EDGE: serger

BEFORE



Finish only free edge

ACTION



Finish only free edge

CHECKED RESULT



Finish only free edge

WHAT TO DO

1. Test the cutout curve on scrap before handling the actual support.
2. Overedge only the free upper edge, including the left flange cutout, with the soft qualified edge finish.
3. Guide the curve without stretching it. Keep the knife from changing the finished opening.
4. Secure the chain tail with a blunt needle through the stitching; keep a sample for comparison.

SEWING OPERATION

Finish free pocket edges - soft three-thread overedge; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Flange opening retains its shape without a tight corded edge.

- Sides and bottom still retain their attachment allowances.

AVOID THIS

Do not finish the cutout with a nonstretch straight seam or sew the opening shut.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

Finish the upper free edge

Parts #2b, #6 | EDGE: serger

BEFORE



Finish only free edge

ACTION



Finish only free edge

CHECKED RESULT



Finish only free edge

WHAT TO DO

1. Lay the upper support in interior orientation: cutout on its lower left.
2. Finish the lower insertion edge and cutout using the same approved soft overedge operation.
3. Control the curve and secure tails without trimming into the flange clearance.
4. Place both supports together and confirm their cutouts face one another.

SEWING OPERATION

Finish free pocket edges - soft three-thread overedge; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Both flange reliefs are on the same diagram side.
- The opposed free edges stay flexible and accessible.

AVOID THIS

A 180-degree rotation moves the cutout to the wrong side; use a vertical reflection.

RECORD

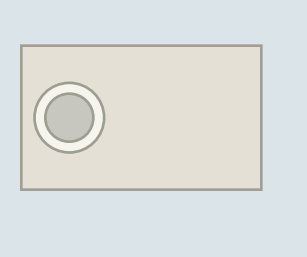
Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

ASSEMBLY / STEP 06

Position the interior supports

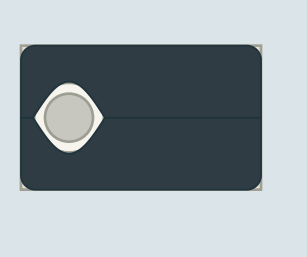
Parts #1, #2a, #2b | POCKET: standard sewing machine

BEFORE



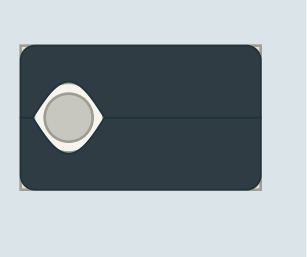
Inside view / left flange

ACTION



Inside view / left flange

CHECKED RESULT



Inside view / left flange

WHAT TO DO

1. Lay #1 wrong side up. Center the bag envelope in the 12-inch pouch zone.
2. Turn under the 0.25-inch attachment allowances on support sides and outer perimeter, keeping free edges unfolded.
3. Place lower and upper supports around the bag envelope; align opposed cutouts around the flange.
4. Clip and mark the attachment perimeter. Remove the bag before any needle enters the work.

SEWING OPERATION

Attach turned pocket perimeter - stretch-capable zigzag; coupon-qualified

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Support width is centered within the expansion zone.
- The facing free edges, flange and outlet remain outside attachment lines.

AVOID THIS

Do not sew the supports to each other across the central insertion opening.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

Baste and check support access

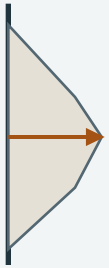
Parts #1, #2a, #2b | POCKET: standard sewing machine

BEFORE



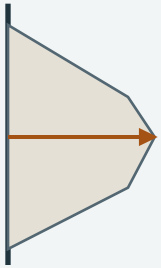
Reference plane

ACTION



Progressive gauge

CHECKED RESULT



Target 1.5 in

WHAT TO DO

1. Baste the turned attachment perimeters through the supports and body only.
2. Insert the horizontal reference bag or mock-up; check insertion, removal and outlet access.
3. Use the depth gauge progressively up to 1.5 inch outward from the local body plane. Check retention without forcing.
4. Remove the reference before sewing. If clearance fails, record the change needed and revise the shared design first.

SEWING OPERATION

Attach turned pocket perimeter - stretch-capable zigzag; coupon-qualified

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- No flange or outlet obstruction at the tested depth.
- Record this as preliminary support fit, not final garment validation.

AVOID THIS

A flat bag fitting does not prove that a filled pouch will fit.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

ASSEMBLY / STEP 08

Attach the lower support

Parts #1, #2a, #6 | POCKET: standard sewing machine

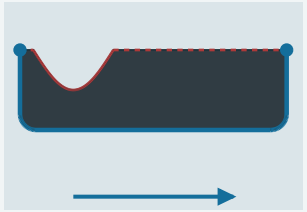
BEFORE



Dots = start / stop

Red dash = leave open

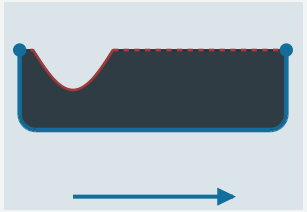
ACTION



Dots = start / stop

Red dash = leave open

CHECKED RESULT



Dots = start / stop

Red dash = leave open

WHAT TO DO

1. Keep #1 open and flat, wrong side up; move the upper support clear.
2. Use the approved stretch zigzag along the turned side and bottom perimeter, about 1/8 inch inside the folded edge.
3. Begin and end at the free-edge junctions. Secure ends with the tested machine method without a bulky dense tack.
4. Remove basting and check that only the intended layers were caught.

SEWING OPERATION

Attach turned pocket perimeter - stretch-capable zigzag; coupon-qualified

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Free top edge and flange cutout remain unstitched to #1.

- Stitching stays on the turned allowance and extends without cracking.

AVOID THIS

Sewing across the free edge blocks insertion and changes the pocket function.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

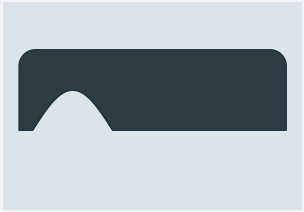


Layer schematic

Attach the upper support

Parts #1, #2b, #6 | POCKET: standard sewing machine

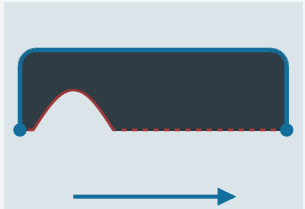
BEFORE



Dots = start / stop

Red dash = leave open

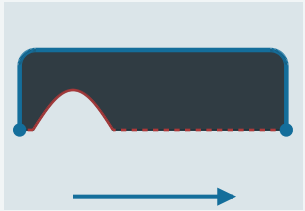
ACTION



Dots = start / stop

Red dash = leave open

CHECKED RESULT



Dots = start / stop

Red dash = leave open

WHAT TO DO

1. Move the lower support clear and retain the marked upper-support position.
2. Attach only the turned sides and upper outer perimeter using the same qualified stretch zigzag.
3. Secure start/stop points, remove basting and inspect the exterior for puckering.
4. Recheck insertion and clearance with both supports attached, then remove the reference bag.

SEWING OPERATION

Attach turned pocket perimeter - stretch-capable zigzag; coupon-qualified

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- The lower edge stays open; left cutouts align.

- Both supports retain the reference without a seam crossing the flange.

AVOID THIS

Do not accidentally stitch the lower support into the upper perimeter.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

ASSEMBLY / STEP 10

Hem and mark the back insert

Parts #4, #6 | HEM: coverstitch

BEFORE



Top / body / bottom
2 / 7 / 2 in

ACTION



Top / body / bottom
2 / 7 / 2 in

CHECKED RESULT



Top / body / bottom
2 / 7 / 2 in

WHAT TO DO

1. Turn the short top and bottom edges by 0.25 inch; coverstitch using the qualified needle spacing and hem coupon.
2. Finish before the long side seams. Do not stretch the narrow insert through the foot.
3. Mark the finished side edge into top 2 / middle 7 / bottom 2-inch sections.
4. Retain 0.25-inch side seam allowances; secure all coverstitch ends per machine manual.

SEWING OPERATION

Finish insert ends / selected allowances - two-needle coverseam; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Finished insert height is 11 inches.

- Marks align with the future body-to-band seam junctions.

AVOID THIS

Do not reduce the insert to body-only height: it must also bridge both bands.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

Fold the two open edge bands

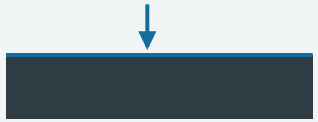
Parts #3 | FOLD: clips / hand positioning

BEFORE



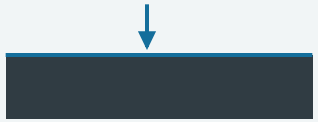
Two open cut strips

ACTION



Fold; keep ends open

CHECKED RESULT



Fold; keep ends open

WHAT TO DO

1. Fold each strip lengthwise, wrong sides together, aligning its two long raw edges.
2. Mark midpoint and the boundaries of the centered 12-inch attachment zone at the seam line.
3. Keep both short ends open with their joining allowances intact. Do not join them into loops.
4. Check the two pieces against each other before attaching either.

SEWING OPERATION

Fold matching edge bands - none; removable holding basting only

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Both strips are identical, with 2-inch finished projection after joining.

- Folded outer edges are clean and raw attachment edges align.

AVOID THIS

The elastic insert supplies the remaining back width; the bands must not bypass it.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

Baste the top band flat

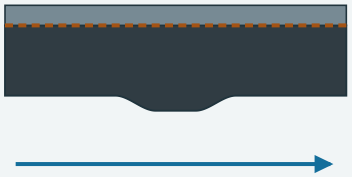
Parts #1, #3 | EASE: hand basting then serger

BEFORE



Exterior schematic

ACTION



Exterior schematic

CHECKED RESULT



Exterior schematic

Check before continuing

WHAT TO DO

1. Place the folded top band on the body right side with raw edges aligned at the upper attachment seam.
2. Match the center and pouch-zone boundaries, then the straight side sections.
3. Distribute the computed 0.639-inch excess only within the shaped zone using fine removable basting.
4. Turn the band outward and inspect the expansion shape; do not trim away the extra material.

SEWING OPERATION

Ease expansion into straight bands - temporary basting then qualified JOIN

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- No hard pleats or severe puckering; compare with the opposite band.

- Stop and revise pattern or material if the excess cannot be distributed cleanly.

AVOID THIS

Do not stretch the entire band or body to hide a local seam-length mismatch.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

ASSEMBLY / STEP 13

Sew the top attachment

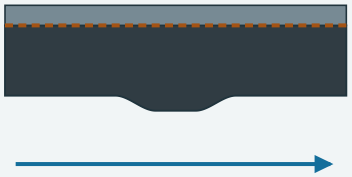
Parts #1, #3, #6 | JOIN: serger

BEFORE



Exterior schematic

ACTION



Exterior schematic

CHECKED RESULT



Exterior schematic

Check before continuing

WHAT TO DO

1. Remove the reference bag. Return the band to the basted joining position.
2. Join with the qualified four-thread overlock at the 0.25-inch allowance, catching body and both band layers.
3. Feed the shaped zone carefully; remove clips before the foot and do not cut the seam line.
4. Turn the band outward, settle allowance toward the body and inspect before removing remaining basting.

SEWING OPERATION

Structural join - four-thread overlock; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Band projects 2 inches from its seam to folded edge.
- The body still has outward ease; both short ends remain open.

AVOID THIS

Do not catch an interior support or free pocket edge in the band seam.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

Baste the bottom and check depth

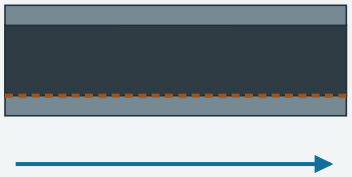
Parts #1, #3 | EASE: hand basting then serger

BEFORE



Exterior schematic

ACTION



Exterior schematic

CHECKED RESULT



Exterior schematic

Check before continuing

WHAT TO DO

1. Repeat alignment for the bottom band, with the same centered easing zone and open ends.
2. Temporarily clip the full-height insert to both assembled ends at the finished side seams.
3. Place on the curved fixture and measure outward depth up to 1.5 inch; also check the top band and supports.
4. Remove the mock-up and temporary insert clips before returning to the machine.

SEWING OPERATION

Ease expansion into straight bands - temporary basting then qualified JOIN

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Reference remains insertable, removable and retained; outlet accessible.

- Record results. Do not permanently sew if any required clearance fails.

AVOID THIS

Do not interpret the 10-inch flat body height as proven outward capacity.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

Sew the bottom attachment

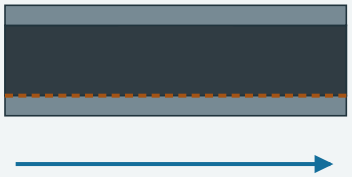
Parts #1, #3, #6 | JOIN: serger

BEFORE



Exterior schematic

ACTION



Exterior schematic

CHECKED RESULT



Exterior schematic

Check before continuing

WHAT TO DO

1. After the basting checkpoint passes, join the bottom band using the same allowance and seam formation as the top.
2. Catch the body and both band layers without catching the interior supports.
3. Turn outward and settle the allowance toward the body without heat that damages elastane.
4. Lay the assembly flat and align both full-height end edges for the back-insert seams.

SEWING OPERATION

Structural join - four-thread overlock; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Top and bottom use identical fabric, folds and finished height.

- Each assembled end matches the insert section marks.

AVOID THIS

An unequal seam allowance makes the two bands finish at different heights.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

ASSEMBLY / STEP 16

Join the first full-height back seam

Parts #1, #3, #4, #6 | JOIN: serger

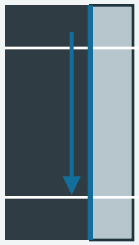
BEFORE



Full-height join

Match both junctions

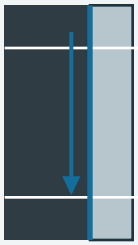
ACTION



Full-height join

Match both junctions

CHECKED RESULT



Full-height join

Match both junctions

WHAT TO DO

1. Place one finished insert edge against one end of the body-and-bands assembly, right sides together.
2. Match the top edge, upper band seam, lower band seam and bottom edge.
3. Sew one continuous qualified overlock seam through top band, body and bottom band at 0.25 inch.
4. Keep the insert relaxed; grade bulk only after confirming seam security on the coupon.

SEWING OPERATION

Structural join - four-thread overlock; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Section joins match top 2, middle 7, bottom 2 inches.
- No hard ridge or skipped stitch at the layer transitions.

AVOID THIS

Joining only the body leaves both edge bands disconnected from the insert.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

Close the second back seam

Parts #1, #3, #4, #6 | JOIN: serger

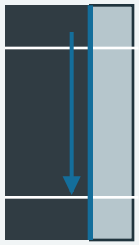
BEFORE



Full-height join

Match both junctions

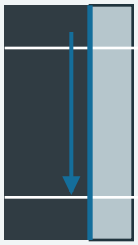
ACTION



Full-height join

Match both junctions

CHECKED RESULT



Full-height join

Match both junctions

WHAT TO DO

1. Bring the remaining assembly end to the opposite insert edge, right sides together.
2. Check that the body and both bands form one untwisted loop. Match all section marks.
3. Sew the second full-height seam, taking care not to catch the opposite side of the loop.
4. Turn right side out and confirm the insert is centered at the back across the full height.

SEWING OPERATION

Structural join - four-thread overlock; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Nominal relaxed seam-line circumference is 36 inches before fabric behavior is evaluated.

- Both bands terminate at the elastic insert rather than closing separately.

AVOID THIS

Stop if a band is twisted; do not solve it by cutting or adding an unrecorded seam.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

Secure and finish the seams

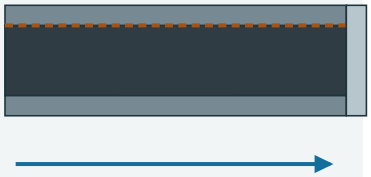
Parts #1, #2a, #2b, #3, #4, #6 | HEM: coverstitch

BEFORE



Exterior schematic

ACTION



Exterior schematic

CHECKED RESULT



Exterior schematic

Check before continuing

WHAT TO DO

1. Secure every overlock and coverstitch end using the approved machine method; trim only loose tails.
2. Inspect side joins and band junctions for needle damage, seam cuts and missed layers.
3. Use a qualified coverstitch only where a sample shows a comfort benefit from holding allowance flat.
4. Keep fold edges and the exterior clean; decoration is optional and must not limit stretch.

SEWING OPERATION

Finish insert ends / selected allowances - two-needle coverseam; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- No unsecured chains or rough knots on the skin-facing surface.

- No dense tack or topstitch restricts the back insert.

AVOID THIS

Do not pull thread tails until stitches unravel; do not bury bulky knots at the flange.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

Inspect access and expansion

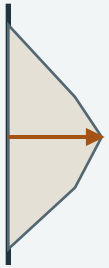
Parts #1, #2a, #2b, #3, #4 | EASE: hand basting then serger

BEFORE



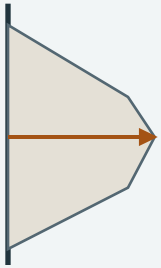
Reference plane

ACTION



Progressive gauge

CHECKED RESULT



Target 1.5 in

WHAT TO DO

1. On the fixture, insert the reference and test intermediate depth before the 1.5-inch target.
2. Measure perpendicular to the local body plane at the fullest part, not from the outer band edge.
3. Check flange/outlet access, retention, recovery and smooth band placement at each stage.
4. Assess required slip-on extension on a suitable fixture; record the limiting material or seam.

SEWING OPERATION

Ease expansion into straight bands - temporary basting then qualified JOIN

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- No broken stitches, blocked opening or loss of retention.

- Any failed function returns the affected pattern and instructions to review.

AVOID THIS

Passing a fixture check is not a measured compression or clinical performance claim.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.

ASSEMBLY / STEP 20

Record the prototype results

Parts #1, #2a, #2b, #3, #4, #6 | HEM: coverstitch

BEFORE	ACTION	CHECKED RESULT
☐ _____	☐ _____	☐ _____
☐ _____	☐ _____	☐ _____
☐ _____	☐ _____	☐ _____
☐ _____	☐ _____	☐ _____
☐ _____	☐ _____	☐ _____
Measure	Document	Review / retest

WHAT TO DO

1. Record fabric lots, machine settings, dimensions, tested extension and all depth-check observations.
2. Launder only by the selected materials care instructions; record actual cycle and drying method.
3. Remeasure after relaxation and repeat access, seam and recovery checks.
4. Attach dated results and photos to the revision. Leave unperformed checks as requires retest.

SEWING OPERATION

Finish insert ends / selected allowances - two-needle coverseam; confirm formation

Joining allowance: 0.25 in where specified. Use the approved coupon settings.

CHECK BEFORE CONTINUING

- Every claimed pass has a dated record and tested configuration.

- Material or geometry changes invalidate the affected result.

AVOID THIS

Do not label an informal wash or stretch check as ISO-compliant testing.

RECORD

Operator / date / coupon ID / pass or issue. Stop and review dependent parts if the check fails.



Layer schematic

RESEARCH APPENDIX

Sources and limits

Source descriptions are paraphrased. Links are live in the SVG index; no standards text is reproduced.

AE / American & Efird - Sewing Stretch Knit Fabrics

Operator feeding, stretch seams and sample troubleshooting.

https://www.amefird.com/wp-content/uploads/2010/01/Sewing-Stretch-Kni-tFabrics-2-6-10.pdf

Accessed 2026-10-14 / manufacturer guidance

COATS / Coats - Soft and Secure Seams

Needle / looper thread roles, skin-side softness and extension.

https://www.coats.com/pl/info-hub/about-soft-and-secure-seams-for-activewear-and-intimates/

Accessed 2026-10-14 / manufacturer guidance

SCHMETZ / SCHMETZ - Stretch Needles

Domestic stretch needle characteristics; use the actual machine needle system.

https://www.schmetzneedles.com/products/stretch-sewing-machine-needles

Accessed 2026-10-14 / manufacturer guidance

PEGASUS / Pegasus - Stitch Library

Stitch formation examples; confirm actual machine formation before assigning codes.

https://www.pegasus.co.jp/stitch/index.php?lang=en

Accessed 2026-10-14 / manufacturer guidance

NO CERTIFICATION CLAIM

Terminology standards do not certify the product. Coupon and laundering checks here are informal prototype checks; full laboratory procedures require the applicable standard and equipment.

RESEARCH APPENDIX

Sources and limits

Source descriptions are paraphrased. Links are live in the SVG index; no standards text is reproduced.

ISO4915 / ISO 4915:1991 - Stitch classification

Terminology reference, not a garment performance certification.

https://www.iso.org/standard/10932.html

Accessed 2026-10-14 / standard catalog

ISO4916 / ISO 4916:1991 - Seam classification

Seam terminology reference, not a compliance claim.

https://www.iso.org/standard/10934.html

Accessed 2026-10-14 / standard catalog

ISO20932 / ISO 20932-1:2018 - Fabric elasticity

Laboratory method reference; informal coupon checks here are not ISO tests.

https://www.iso.org/standard/69489.html

Accessed 2026-10-14 / standard catalog

ISO6330 / ISO 6330:2021 - Washing and drying

Controlled laundering reference; record the actual prototype care process.

https://www.iso.org/standard/75934.html

Accessed 2026-10-14 / standard catalog

NO CERTIFICATION CLAIM

Terminology standards do not certify the product. Coupon and laundering checks here are informal prototype checks; full laboratory procedures require the applicable standard and equipment.

PROTOTYPE WORKSHEET

Setup And Measurements

Complete by hand or in the source-controlled physical test record. Blank fields are not passed tests.

Operator / date / sample ID / revision

Material supplier, lot and stretch direction for each part

Machine model / needle system / needle size

Needle threads / looper threads / bobbin thread

Stitch formation / length / width / tension / differential feed

Approved coupon IDs and required extension

Relaxed circumference / side height / folded-band heights

Actual wash / dry procedure and before-after measurements

PROTOTYPE WORKSHEET

Fit, Expansion And Change Record

Complete by hand or in the source-controlled physical test record. Blank fields are not passed tests.

Fixture and bag/mock-up description; flange and outlet dimensions

Depth measured from local body plane; intermediate / target result

Insertion / removal / flange and outlet clearance

Retention / band flatness / recovery / seam defects

Slip-on extension tested and limiting seam or material

Photograph / measurement record locations

Change requested; other affected parts and operations

Pass / fail / not evaluated; next revision / retest required